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DUAL MODULATION OF CD44 P-GP PATHWAYS USINGHYALURONIC ACID NANOCARRIERS FOR COLON CANCER THERAPY

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ABSTRACT:

Colon cancer stands as one of the main reasons people die from cancer around the world because multidrug resistance (MDR) creates treatment challenges for patients. The overexpression of P-glycoprotein (P-gp) serves as the main cause of this resistance because P-glycoprotein functions as an ATP dependent efflux transporter which restricts drug buildup inside cells. The transmembrane glycoprotein CD44 exists in colon cancer cells and cancer stem cells at increased levels, which enables cancer cells to spread throughout the body while developing resistance to treatment. The hyaluronic acid-based drug delivery system has developed into an effective method for targeting CD44 positive colon cancer cells because it uses receptor-mediated endocytosis to improve drug bioavailability. The therapeutic effectiveness of treatments remains blocked by P-gp activity. The study investigates how HA-based CD44 targeting works together with P-gp inhibition to fight multidrug resistance in colon cancer treatment. The study examines HA and CD44 structural attributes and functional properties while explaining how P-gp functions as a drug expulsion mechanism and establishes the design of HA-conjugated nanoparticles for precise and continuous drug distribution. Emphasis is placed on recent developments in HA-P-gp inhibitors nanocarriers, their preclinical and clinical evidence, and the prospects of integrating these systems into personalized medicine framework. The combination of HA-based targeting and P-gp modulation represents a novel and rational therapeutic strategy for enhancing the sensitivity of colon cancer cells to chemotherapeutic agent while minimizing systemic toxicity.

Keywords: *Hyaluronic acid, CD44, P-glycoprotein, Multidrug resistance, Colon cancer, Nanoparticle, Targeted drug delivery.*

Introduction

In the US,Colorectal cancer (CRC) is currently the third most prevalent cancer diagnosis for both men and women (Marley & Nan, 2016).Nowdays, more people are being diagnosed with colorectal cancer (CRC), sometimes at an early stage and sometimes at an advanced stage,thanks to the widespread availability of colonscopy.The world Health Organization (WHO) reports that 1.80 million new instances of colorectal cancer are diagonised worldwide in 2018, and 862,000 persons lost their lives to the disease.Every year, about 145,600 instances of colorectal cancer are diagnosed in the united states.According to the (Siegel et al., 2019) ,there are 1,014,200 instances of colon cancer and the remaining cases are rectal cancer.In general, the colon acconts for 71% of CRC cases, whereas the rectum accounts for 29%. Every year in the USA, almost 50,000 people pass away from CRC.The incidence of colorectal cancer (CRC) declined somewhat in men (2.2% annually) but stayed constant in women between 2008 and 2014. Between 1999 and 2015 , the death rate from colorectal cancer (CRC) ALSO DRPPED BY 1.8% annually for males and 1.4% annually for women (Cronin et al., 2018). UP to 25% of instances occur in people with a family history of colorectal cancer, and around 10% of cases occur in hereditary colorectal cancer syndromes, despite the fact that the majority of CRC cases (amost 70%) occur in average risk persons (Hadjipetrou et al., 2017). Based on genetic research, we now have a much better understanding of the pathophysilogy of both hereditary and sporadic colorectal cancer (Weitz et al., 2003). Targeted treatments and checkpoint inhibitors are selected for metastatic colorectal cancer (CRC) based on the presence of K-ras mutations, mismatch repair(MMR) gene defects, amd microsatellite instability (MSI).The non- kinase transmembrane glycoprotein CD44 is overexpressed in a number of cell types,including cancer srem cells,and frequently exhibits alternative spliced variants that may be involved in the initiation and spread of cancer.The primary ligand for CD44, hyaluron,binds to and activates CD44 ,triggering cell signalling pathways that promote cell proliferation,boost cell survival,alter cytoskeletal alteration and improve cellular motility.Uncertainity surrounds the distict functions of CD44 standard(CD44s)and specific CD44 variant (CD44v) isoforms.Additional peptide motifs found inCD44v can interact with and sequester cytokines and growth factors at the cell surface,acting as a corecptors to support cell signalling .Additionally, CD44v was expressed in tumours that had spread ,whereas alternating between CD44v andCD44s may be important for controlling the epithelial to mesenchymal transition(EMT) and the adaptability of cancer cells.Here ,we examine recent findings about CD44s structural and functional characterstics, its recognized roles in carcinogenicity , the control of CD44 expression , and the possibility of using CD44 as a target for cancer treatment.According to ,(Gronthos et al., 2001), the CD44 family of non-kinase ,single pan transmemberane glycoproteins is expressed on embryonic stem cells and at varying amounts

on other cell types, such as bone marrow and connective tissues.(Domev et al, 2012)CD44 expression is known as a molecular marker for cancer stem cells(CSC) and is elevated in cancer cell subpopulations. (Yin et al., 2016).Ten of the 19 exons that make up the human CD44 gene are consistent throughout all isoforms.The ten constant exons and any combination of the remaining nine variable exons make up CD44v variant isoforms 9cd44V),which are produced by alternative splicing (Screaton et al., 1993), (Prochazka et al., 2014).Hyaluronic acid (HA),a plentiful extracellular matrix(ECM) component expressed by cancer and stromal cells, is the primary ligand for cd44 (Banerjee et al., 2016).When HA binds to the CD44 ligand binding domain, it causes conformational changes that enable adaptor proteins or cytoskeletal components to bind to intracellular domains.These changes then trigger a variety of signalling pathways that result in cell adhesion,invasion,migration,and proliferation(Ponta et al., 2003)(Zöller, 2011) . The functions of CD44s and ifferent are both similar and different.Additional binding motifs found in CD44v isoforms facilitate CD44s interaction with molecules in the microenvironment (Yamano et al., 2016).By binding or sequestering growth factors on the cell surface and delivering them to their particular receptors,CD44v,isoforms can function as co-recptors (Orian-Rousseau et al., 2002).When cancer cells goo through an epithelial to mesenchmal transition(EMT), they become more like stem cells and express mpre CD44 (Mani et al., 2008).Additionally, cancer cells that exhibit an EMT phenotype are more invasive and chemotherapy - resistant (Zhao et al., 2016).CD44 may be a molecular target for cancer treatment based on the clinicopathological effects of CD44S and its isoforms in promoting carcinogenesis.Furthermore,CD44 may be a significant prognostic marker due to its known role in preserving stemness and the significance of cancer stem cells in tumour regeneration after therapy.Clinical development of therapeutic approaches that target CD44 or lower CD44 expression is in different stages (L. Li et al., 2014),(Matzke-Ogi et al., 2016),(Todaro et al., 2014).These tactics include aptamers, ectodomain mimics,tumour delivery of shRNAs, and CD44 neutralising antibodies (Murphy et al., 2005),(Orian-Rousseau & Ponta, 2015),(Iida et al., 2014).

2. Hyaluronic acid (HA) and CD44 Interaction

Through signalling networks such RhoGTPases and the PI3/AKT patway (Okamoto et al., 2001) as well as other chemoresistance pathways via ROK activation,HA's association with the CD44 receptor promotes cell survival,proliferation,invasion,and metastasis.The CD44 receptor promotes tumour spread by triggering cell survival. When the HA ligand binds to CD44,it activates the cytoplasmic domain of CD44, which causes a sequence of cell signalling events and the recruitment of different proteins.Ankyrin,Merlin,and ERM are examples of CD44-Cytoplasmic domain activated proteins that participate in actin polymerisation,which is

mediated by ERM proteins and leads to cytoskeleton rearrangements for tumour invasion and migration.(Bretscher et al., 2002)

In a number of tumors,CD44 intteraction is linked to metastasis.(Yu & Stamenkovic, 2000),(Yu & Stamenkovic, 1999)Although CD44 participation in triggering apoptosis prevents tumour invasion and progression,its significance in cancer progression is extremely evident.(Deed et al., 1997)and(Naor et al., 2002),CD44 is HAMW reliable in causing apoptosis and decreasing cell invasion, but it is still inconsistent,indicating that the response varies in different tissues as well,In breast colon and mammary cancer cells, the high molecular weight HA prevented invasion,where as the tiny molecular weight oligomerized HA started apoptosis and inhibited the activation of the activation of the cell survival and growth pathway(AKT)(Lopez et al., 2005)(Ghatak et al., 2002)

In addition to AKT, RhoGTPases initiate a series of events that include downstream signals and cell surface receptors.RhoGTPases function as reaction molecules that react to signals or external stimuli and control different cytoskeleton reorganization processes that lead to cell invasion and metastasis.RhoGTPases, which comprise RhoA,Rac1,and Cdc42, are members of the ras superfamily.When GTP molecules attach to these Rho proteins,they change form an inactive to an active state.Normal cell invasion and development are facilitated by the RhoGTPases . Nonethless their overexpression in a number of human maliganancies is associated with poor progonosis and disease progression.(Bourguignon et al., 2008) Furthermore, a crucial kinase called Rho kinase(ROK)controls the invasion and metastasis of tumours through myosin light chain phosphatase.Myosin light chain phosphates stimulates the release of matrix metalloproteins9MMPs),which are proteins that break down the extracellular matrix.By breaking down collagen,the activation of mmps through the CD44 receptor protein aids in both invasion and angiogenisis.(W. T. Chen, 1996)(Gum et al., 1996) The MMPs,which are essential for tumour invasion and metastases,include mmp-2(typefour collagenase),mmp-9(typefive collagenase),mmp-3(stromelysins), and mmp-1(interstitial fibroblast – type collagenase)

Through crucial RhoA signaling the HA/CD44 connection enables cytoskeleton activation.RhoA modifies intracellular Ca levels to control cellular processes(Schaefer & Schaefer, 2010).Essential phospholipase c (PLC) is responsible for the intracellular mobilization of calcium(Schaefer & Schaefer, 2010).Ca is released from deposit storage when plcs are activated and transformed to inositol triphosphate (IP3) via pip2 (hydrolysed).According to Schaefer and Schaefer (2010),this released ca promotes cytoskeleton reorganization and ultimately cancer growth(Bourguignon et al., 2006)(Singleton & Bourguignon, 2002) HA stimulation of CD44 in brest cancer cells 107-109 causes the production of the anti-apoptotic gene Bcl and the

multidrug resistance gene p- glycoprotein,both of which support the growth and survival of tumour cells..(Lakshman et al., 2004)(Park et al., 2012) The highly controlled process of cell death known as apoptosis is marked by morphological and biochemical alterations, including a number of protreins and enzymes that both cause and sustain cell development.(Jan & Chaudhry, 2019)

However, Bourguignon 2019 recently proposed that HA-CD44 enhanced tumour growth signaling events incancer.(Bourguignon, 2019) Tumour cell survival,proliferation,invasion and migration are all impacted by the interaction and binding HA to CD44v, which also increases the expression of Ras/Cdc42/Rac1/RhoA/Nfkb and other cytokines. As previously mentioned,pI3k/AKT activation and the synthesis of LncRNA9(UCA1)are both stimulated by HA-AKT activation.Via ROK, LncRNA induced the migration and invasion of cancer cells.Additionally, HA/CD44 causes chemoresistance by activating the production of miR-10b,miRNA-21, and miR302 that are activated by transcriptional factors.Through phosphorylation, JNK activity activates c – jun.The up – regulation of miR-21 expression is caused by thr activated c- jun binding to the miR-21 promoter.Similarly, the complex assembly of Nanog, Stat – 3,DROSHA,and p68 is triggered by the HA- CD44 interaction.This complex binds to the miR-21 Promoter region, causing miR-21 to be expressed.IAP protein, which is implicated in chemoresistance and cancer cell migration and invasion are stimulated by activated miR-21. On the other and,because of the HA-CD44 connection, enhanced miRNA-302 production also contributes to chemoresistance .HA binding to CD44 facilitates the recruitment and formation of a connection between OCT4/SOX2/Nanog and the cytoplasmic region of CD44v3.Through the evaluation of miR-302 gene expression, this OCT/SOX2/Nanog induces IAP(clap-1),clap-2, and XIAP) expression.IAP contribute to the development and chemoresistance of cancer cells.Furthermore, the HA-CD44 interaction triggers the cytoskeleton activation caused by MiRNA-10b expression. Twist is phosphorylated by active c-src, which is facilitated by HA binding to CD44.Tumour cell movement and invasion are encouraged by the active Twist, which also function as TF and activates mR-10b expression.

2.1 Structure and properties of Hyaluronic acid

HA belongs to the glycosaminoglycan group of heteropoly saccharides. Unlike other Glycosaminoglycans (i.e., chondroitin sulfate, dermatan sulfate, keratin sulfate, heparin Sulfate, and heparin), HA is not sulfated and is not synthesized by Golgi enzymes; thus, it Is not covalently bound to the core protein(Fallacara et al., 2018),(Gottschalk & Elling, 2021). Synthesis by HASs takes place on the Cytoplasmic surface of the plasma membrane, and the HA is usually directly extruded Through plasma membranes into the extracellular space (Lierova et al., 2022). HA also differs in size, as it can Reach up to 10^4kDa, whereas other glycosaminoglycans range from 15 to 20kDa (Fallacara et al., 2018). It is an Unbranched linear polysaccharide composed of

repeating disaccharide units, which consist Int. of GlcNAc and GlcA linked by alternating β-1,4 and β-1,3 glycosidic bonds (Figure 1)(Yao et al., 2021)(Y. Wu et al., 2024). In the β-conformation, hydroxyl, carboxyl, acetamido, and anomer carbons are in sterically Favorable equatorial positions, making polysaccharide chains very energy-stable. The Hydrogens occupy less favorable axial positions. In this rigid conformation, with limited Rotation around the glycosidic bonds of the HA backbone, hydrophobic groups alternate With polar groups linked by intra- and inter-molecular hydrogen (Fallacara et al., 2018)]. At physiological pH, each carboxyl group carries an anionic charge, which can interact with mobile cations and form salts called hyaluronan or hyaluronate(Fallacara et al., 2018)(S. Zhang et al., 2023)]. HA is highly hydrophilic due to the many carboxyl and hydroxyl groups in its molecule, allowing it to hold approximately 1000 times its weight in water(Jabbari et al., 2023). Water molecules stabilize the secondary structure of HA, a Two-fold single helix, as they link HA carboxyl and acetamido groups with hydrogen bonds. These two-fold helices form in aqueous solution duplexes, i.e., a β-sheet tertiary structure, Formed by hydrophobic interactions between hydrophobic regions and intermolecular Hydrogen bonding between antiparallel HA chains, enabling the aggregation of polymeric Chains(Fallacara et al., 2018),(Y. Wu et al., 2024). The establishment of polymeric networks is dependent on HA's molecular Weight and concentration, as an increase in both parameters results in a strengthened Network, as well as increased viscosity and viscoelasticity. HA solutions behave as nonNewtonian fluids with viscoelastic and shear-thinning (pseudoplastic) characteristics. The Rheology of HA is strongly influenced by the ionic strength of the solution, pH, and Temperature. At pH values below 4 or above 11, hydrolytic degradation of HA occurs, Resulting in disintegration of the polymer network and reduced viscosity (Fallacara et al., 2018),(S. Zhang et al., 2023). Under Acidic conditions, the proton attacks the GlcA unit, while the alkaline hydroxide ion (Fallacara et al., 2018)attacks The GlcNAc(J. Li et al., 2020). The polymer network's breakdown and shift from a gel to a liquid is Transient and reversible (Fallacara et al., 2018). The unique properties of HA are crucial for its physiological Roles and its applications in various industrial fields(Fallacara et al., 2018),(Y. Wu et al., 2024).

2.2 CD44 receptor: biology and overexpression in colon cancer

CD44 is a transmembrane multifunctional glycoprotein that is a member of the cellular adhesion molecule (CAM) superfamily, which has been studied extensively for its role in cell–cell and cell–matrix adhesion. CD44 is composed of three functional domains: an extracellular hyaluronic acid (HA) binding domain, a transmembrane domain that anchors CD44 within the cell membrane, and a short cytoplasmic tail that allows interaction with cytoskeletal proteins and signaling molecules. The gene for CD44 undergoes alternative splicing, producing different isoforms of CD44, beginning with a standard non-variable isoform (CD44s) and

producing many variant isoforms (CD44v), where the difference among isoforms is mainly located in the extracellular region where individual variant exons are included.(Ziranu et al., 2024) The molecular diversity of CD44 also enables this protein to act as a receptor for several ligands including hyaluronic acid, osteopontin, laminin, collagens, chondroitin sulfate, and growth factors. This diversity underscores the functional versatility of CD44.

In terms of colorectal cancer (CRC), CD44 is a key validated marker for colon cancer stem cells (CSCs). CD44+ CSCs are resilient tumorigenic subpopulations that can initiate tumor growth, sustain tumor growth and facilitate invasion, thereby enhancing relapse and metastasis. Findings from experimental literature indicate that CD44 expression supports critical CSC features including self-renewal, long-term clonogenicity, and conventional cytotoxic drug resistance. Furthermore, CD44 overexpression is correlated with an enhanced ability for epithelial-mesenchymal transition (EMT), which is a biological transition that enables colon cancer cells to transition to more invasive and motile behavior resulting in more rapid metastasis.(Lamouille et al., 2014)

The CD44 variant isoforms (CD44v6, CD44v3 and CD44v8-10), are specifically upregulated in colorectal tumors, and correlated with a higher tumor stage, lymph node metastases and distant organ metastases (particularly liver metastases).(K.-L. Chen et al., 2020) CD44 variants not only act as a passive marker of tumor burden, but further promote tumor progression by stabilizing receptor tyrosine kinases (RTKs), including EGFR, c-Met and VEGFR, on the surface of the colon cancer cell, thus promoting downstream oncogenic signalling.(Talmadge & Fidler, 2010) CD44 signalling is embedded within many malignant signalling pathways including Wnt/β-catenin, RAS-MAPK, PI3K-Akt-mTOR, TGF-β/Smad and the Rho-family GTPases.

Consequently, CD44 serves as a canonical signalling platform that regulates cell growth, survival, differentiation, and cytoskeletal remodelling.Significantly, CD44 overexpression is correlated with adverse treatment outcome in colon cancer. CD44 increases resistance to standard chemotherapeutics, such as 5-fluoruouracil and irinotecan, through the regulation of ATP-binding cassette (ABC) transporters, such as P-glycoprotein (P-gp), which actively efflux anticancer drugs from within cells.(Guo et al., 2022) Thus, CD44 plays an important functional role in the development of multidrug resistance (MDR) and is thought to be a primary barrier in the successful eradication of colon cancer.

2.3.HA as a natural ligand for CD44

Hyaluronic acid (HA) is a common, linear, non-sulfated polysaccharide that serves as a large structural component of the extracellular matrix (ECM) while also playing important physiological roles in tissue hydration, elasticity, and cellular homeostasis. The polymeric backbone formed from repeating disaccharide units of D-glucuronic acid and N-acetyl-D-glucosamine confers an extremely high negative charge density and biochemical character that allows HA to have interactions with many proteins of the ECM and with cellular receptors. CD44 is the primary receptor and most-studied receptor with a high affinity for HA.(Chaudhry et al., 2021) The N-terminal hyaluronan-binding domain (HABD) of CD44 recognizes HA motifs in a manner that can depend on size and conformation, with high-molecular-weight HA (HMW-HA) inducing mostly anti-inflammatory, homeostatic functions, while often fragmented low-molecular-weight HA (LMW-HA) that can be generated via enzymatic activity by hyaluronidases and reactive oxygen species in tumors provides potent signaling that can trigger pro-tumorigenic pathways.(Zhong et al., 2019) HA binding to CD44 results in conformational changes and clustering of the receptor, allowing for association with cytoskeletal adaptor proteins (ezrin, radixin, moesin), lipid rafts, and a host of other intracellular signaling mediators.(Misra et al., 2015)The overall process of HA binding to CD44 and the associated activation of intracellular signalling pathways is schematically represented in Figure 1.

Numerous cellular function, such as proliferation,adhesion,motility,differentiation,and survival, are regulated by this HA-CD44 interaction, which function as a molecular switch. Wnt/B- Catenin, pi3k/AKT, MAPK/ERK, Rho/Rac1 GTPases, and TGF-B- mediated epithelial-mesenchymal transition (EMT) are among the oncogenic pathways that are more activated in colon cancer as a result of CD44 overexpression and higher HA turnover. Together, the signaling pathways enable cancer stem cells ability to survive cell death,proliferate through metastasis,form new blood arteries and fibrous tissue,and evade immune detection.

P-glycoprotein is the main ATP- binding cassette transporter that is expressed more frequently as a result of the interaction between HA and CD44.This process causes multidrug resistance while accelerating the bodys removal of drugs.This characteristic is exclusive to colorectal cancers. HA has become a very promising ligand for targeted drug delivery systems because of of its intrinsic biocompatibility and natural,high – specificity affinity for CD44 .This allows receptor- mediated endocytosis of HA- functionalized nanoparticles and promotes selective therapeutic accumulation in colon cancer cells that overexpress CD44. As a result, the HA-CD44 axis is a crucial molecular connection that cintrolstumor development and holds important translational prospects for precision oncology.

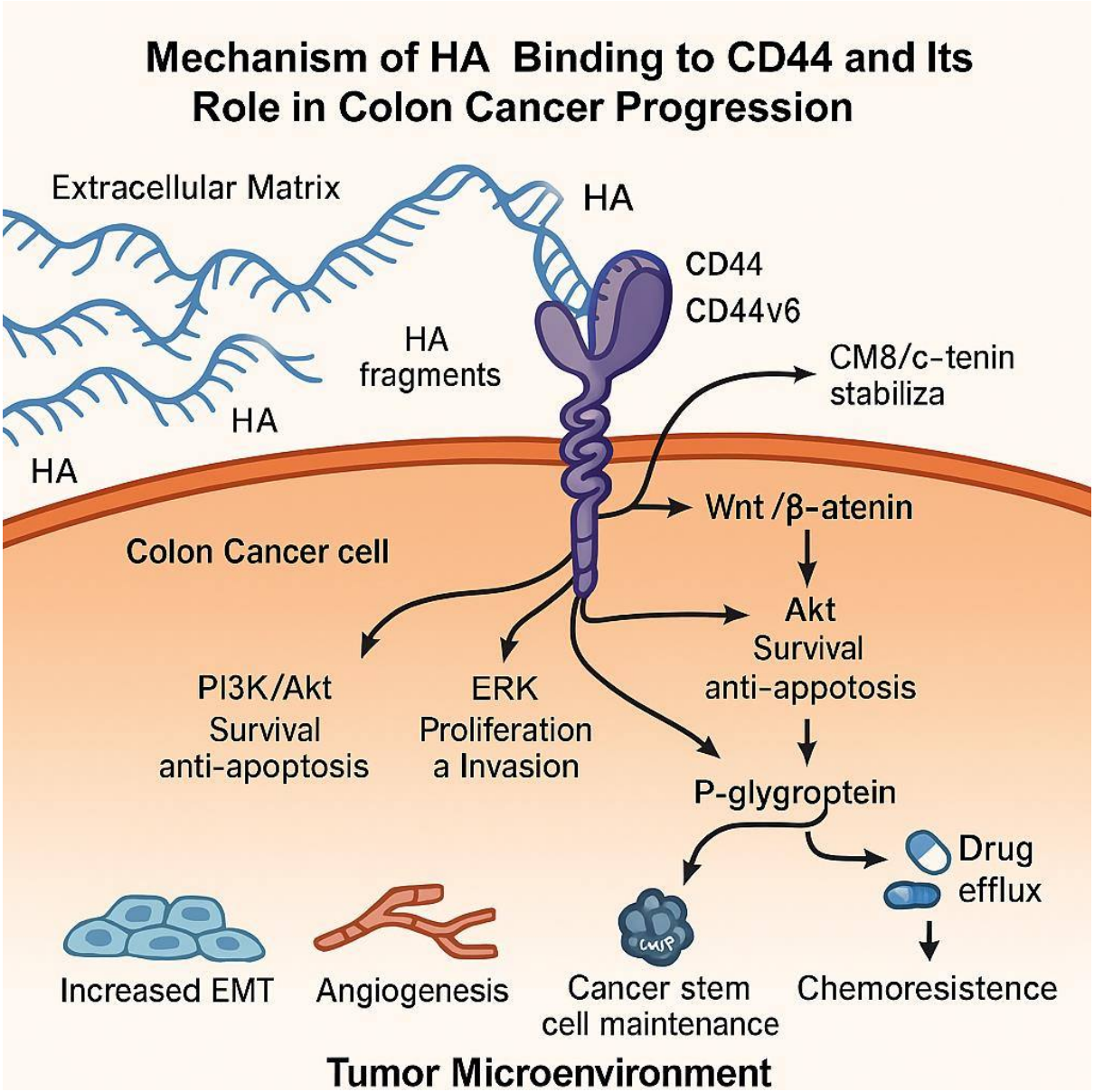


Figure 1 mechanism of HA binding to CD44 and its role in colon cancer progression

3. P-glycoprotein (P-gp) and Multidrug Resistance (MDR)

3.1 Mechanism of P-gp mediated drug efflux

P-glycoprotein (P-gp), also known as ABCB1 or MDR1, is a transmembrane efflux pump belonging to the ATP-binding cassette (ABC) transporter superfamily and is one of the most extensively characterized mediators of multidrug resistance (MDR) in cancer. Structurally, P-gp consists of two homologous halves, each comprising a transmembrane domain (TMD) that forms a drug-binding pocket and a cytoplasmic nucleotide-binding domain (NBD) responsible for ATP hydrolysis.(Tian et al., 2023) The efflux mechanism begins when hydrophobic or amphipathic chemotherapeutic drugs diffuse into the lipid bilayer or cytoplasm and bind to the large, flexible, polyspecific substrate-binding cavity within the TMDs of P-gp. Once the substrate is bound, ATP molecules bind to both NBDs, triggering dimerization of these domains and inducing a significant conformational change in the transporter. This ATP-driven conformational shift reorients the drug-binding pocket from an inward-facing to an outward-facing configuration, thereby expelling the substrate from the intracellular space to the extracellular milieu. The NBD dimer separates from each other after ATP hydrolysis which follows drug extrusion. The process resets P-glycoprotein to its inward-facing configuration with low binding capacity which enables the protein to begin another round of transporting substances. The drug transporter P-glycoprotein reduces drug retention inside cells through its continuous operation which maintains drug levels at less than therapeutic thresholds. This mechanism protects cells from the toxic effects of chemotherapy drugs which include doxorubicin and paclitaxel and irinotecan and 5-fluorouracil metabolites. P-glycoprotein shows high expression levels in colon cancer cases particularly within tumor areas which have high CD44 expression and cancer stem cell populations because hyaluronic acid–CD44 signaling pathways and PI3K/Akt and Wnt/β-catenin pathways increase its expression levels through these pathways.(Mechetner & Roninson, 1992)The coordinated increase of P-gp activity results in strong multidrug resistance phenotypes which cause repeated tumor growth and make standard chemotherapy treatments ineffective. P-gp functions as the main drug resistance mechanism which researchers currently study through P-gp inhibitors and gene silencing methods and CD44-targeted nanocarriers.(Waghray & Zhang, 2018)

3.2 Role of P-gp in Chemotherapy Resistance

P-glycoprotein (P-gp) functions as a key factor which establishes its role in developing chemotherapy resistance across various solid tumors which include colon cancer. P-gp operates as an ATP-dependent efflux pump which the MDR1/ABCB1 gene encodes to expel various chemotherapy drugs from cancer cells. This process results in decreased drug levels inside cancer cells which then protects the cells from experiencing

toxic effects of the drugs. The efflux mechanism prevents many primary anticancer medications from reaching their targets because it affects drugs such as paclitaxel doxorubicin vincristine irinotecan and mitoxantrone which creates both innate and developed resistance to drugs.(Tian et al., 2023)In colon cancer, P-gp shows increased expression in aggressive tumor subpopulations which include cancer stem cells and CD44-overexpressing niche cells that can withstand chemotherapy while reestablishing tumors following treatment. P-gp decreases drug levels inside cells while it creates a multidimensional resistance pattern because it prevents cell death and increases DNA repair abilities and it interferes with oxidative stress management and it interacts with cell survival pathways which include PI3K/Akt and NF-κB and Wnt/β-catenin. The tumor microenvironment which includes hypoxia and inflammatory cytokines and HA–CD44 signaling, increases P-gp expression to enhance the development of resistant traits.(Karthika et al., 2022)P-glycoprotein(P-gp) drug efflux functions as the main barrier which prevents effective chemotherapy treatment resulting in treatment failures and tumor reoccurrence and cancer spread. Researchers continue to investigate P-glycoprotein(P-gp) as a drug target because it plays a key role in multidrug resistance(MDR) and they use P-gp inhibitors and RNA interference and CRISPR gene suppression and ligand-directed nanocarriers to develop better chemotherapy treatment methods for colon cancer patients.(Safar., 2023)

3.3 Clinical Relevance of P-gp in Colon Cancer

The clinical importance of P-glycoprotein shows itself in colon cancer because its increased expression leads to worse treatment results and treatment failures and more frequent tumor recurrences. Clinical studies show that colon tumors with high P-gp levels in their tissues show decreased response rates to standard chemotherapeutic drugs that include 5-fluorouracil and irinotecan and oxaliplatin. The standard treatment regimens FOLFOX and FOLFIRI depend on these drugs but high P-gp activity prevents drugs from entering cells at proper levels which reduces their ability to kill cancer cells. Cancer stem cell populations and CD44-high tumor subregions which promote metastatic spread and post-therapy relapse demonstrate increased P-gp expression(Lin & Yamazaki, 2003). P-gp functions as a clinically useful biomarker which doctors use to assess both the cancer outlook and tumor growth rate and treatment resistance of their colon cancer patients. The presence of hypoxia and inflammation together with hyaluronic acid (HA)–CD44 signaling pathways in the tumor microenvironment activates P-gp expression which develops adaptive resistance for chemotherapy treatment. The presence of high P-gp levels results in both decreased overall survival and disease-free survival rates, which demonstrates its ability to predict patient outcomes. The clinical importance of P-gp extends to treatment strategy development because researchers are testing P-gp inhibitors and HA-based targeted nanocarriers and gene-silencing methods to reduce P-gp effects while increasing drug effectiveness. P-gp

functions as an operational mechanism which causes multidrug resistance while also serving as an essential medical factor that determines how colon cancer patients will respond to treatment and how long they will live and what personalized treatments they will receive.(Zhou, 2008)

4. Hyaluronic Acid-Based Drug Delivery Systems

4.1 HA-Conjugates and HA-Modified Nanoparticles

Hyaluronic acid (HA)-based drug delivery systems have received considerable research interest because of HA's natural ability to work safely with living organisms and its capacity to break down and its water-attracting properties and its strong ability to bind with CD44 receptors that are commonly found in colon cancer cells.(Kim et al., 2019)The therapeutic molecules are attached non covalently or covalently to the HA backbone via chemical groups, including carboxylic, hydroxilic, and N-acetyl.(Kłodzińska & Nielsen, 2021)The HA-drug conjugates display better water solubility and longer circulation time and higher tumor accumulation because they use both passive targeting through the EPR effect and active CD44-mediated uptake. The HA conjugation system protects drugs from initial degradation while establishing controlled release and decreasing toxic effects on the body. The field of drug delivery research has developed HA-modified nanoparticles which include polymeric nanoparticles and liposomes and micelles and dendrimers and nanogels and inorganic nano-formulations to achieve selective drug delivery through receptor-mediated endocytosis(Arpicco et al., 2014). Figure 2 presents a summary of the various HA-based nanocarriers which have been developed for CD44-positive colon cancer treatment through their distinct structural characteristics and functional operational methods and their specific targeting systems. Table 1 presents an overview of the primary HA-based nanocarrier categories which are used in CD44-targeted colon cancer treatment to show their drug contents and operational methods and medical benefits.

HA surface functionalization of nanoparticles leads to better tumor penetration results while enabling cancer stem cells and CD44-overexpressing colon cancer cells to preferentially take up the nanoparticles.(Wei et al., 2013)HA-coated nanocarriers can carry various treatment substances which include chemotherapeutics and P-gp inhibitors and siRNA and gene-editing molecules thus providing a flexible system for combination treatment.(G. Zhang et al., 2025)The tumor microenvironment contains hyaluronidase enzymes which enable HA to undergo targeted degradation that releases drugs specifically at the tumor site. The combination of HA-conjugates and HA-modified nanoparticles creates an advanced targeted delivery system which improves treatment accuracy while overcoming multiple drug resistance and enhancing pharmacokinetics and pharmacodynamics for colon cancer therapy(Hlaing et al., 2022).

Table 1Various hyaluronic acid (HA)-based nanocarriers for CD44-targeted colon cancer therapy

HA-Based Nanocarrier Type	Loaded Drug / P-gp Inhibitor / Gene	Mechanism / Targeting Feature	Key Advantages	References
HA–Polymeric Nanoparticles (PLGA, PEG-PLGA, PCL)	Doxorubicin, Paclitaxel, Tariquidar	CD44-mediated endocytosis; controlled drug release	High drug loading, good biocompatibility, MDR reversal	(Yadav et al., 2007)(Patil et al., 2009)
HA–Liposomes	Irinotecan, SN38, Verapamil	HA-coated lipid bilayers for CD44 targeting	Extended circulation, improved stability, reduced toxicity	(C. Wu et al., 2019)
HA–Micelles (Amphiphilic HA)	5-FU, Curcumin, MDR modulators	Self-assembly; hydrophobic core; CD44 recognition	High solubility for hydrophobic drugs, tumor penetration	(Yuan et al., 2014)
HA–Drug Conjugates	HA–Doxorubicin, HA–SN38, HA–Camptothecin	Covalent HA–drug linkage; hyaluronidase-responsive release	Enzyme-triggered drug release; selective tumor accumulation	(Camacho et al., 2015)
HA–Nanogels / Hydrogel Nanoparticles	siRNA + Doxorubicin; siRNA against MDR1	CD44-targeted gene/drug co-delivery	Excellent water retention; co-delivery of gene + drug	(Ashrafizadeh et al., 2021)(X. Yang et al., 2015)
HA–Mesoporous Silica Nanoparticles (HA–MSNs)	Irinotecan, Oxaliplatin	HA on silica surface; pH-sensitive release	High surface area; deep tumor penetration	(Fatemi et al., 2024)
HA–Gold	DOX or	CD44-mediated	Dual chemo-	(L.-S. Li et al.,

Nanoparticles (HA–AuNPs)	Photothermal Agents	uptake + photothermal therapy	photothermal therapy; imaging capability	2021)
HA–Iron Oxide Nanoparticles (HA–Fe3O4 NPs)	Doxorubicin / siRNA	Magnetic targeting + CD44 targeting	MRI imaging + therapy ("theranostic")	(Kievit et al., 2011)
HA–Carbon Nanotube Conjugates	5-FU, DOX	HA bound to CNT surfaces to reduce toxicity	High loading, enhanced uptake, reduced side effects	(Kearns et al., 2021)
HA–Solid Lipid Nanoparticles (HA-SLNs)	Oxaliplatin, Topotecan	HA-modified lipid matrix	Improved stability; sustained release; CD44 targeting	(German-Cortés et al., 2023)
HA–Metal–Organic Frameworks (MOFs)	Doxorubicin, siRNA	CD44-mediated uptake; pH-responsive release	High loading capacity; tunable pore size	(N. Yang et al., 2025)(Cai et al., 2025)
HA–Exosome-Mimetic Vesicles	Chemotherapeutics or nucleic acids	HA-functionalized biological vesicles	Excellent biocompatibility and natural targeting	(Gebeyehu et al., 2021)(McNamara et al., 2022)

4.2 HA in Targeted Therapy for Colon Cancer

Hyaluronic acid (HA) has developed into a powerful targeting ligand which doctors use for treating colon cancer because of its natural ability to bind to CD44 receptors which colorectal tumors and cancer stem cells display in high amounts. HA-mediated targeting offers dual advantages: it enables selective delivery of therapeutic agents to CD44-positive cancer cells while simultaneously facilitating efficient internalization via receptor-mediated endocytosis. This makes HA an ideal ligand for precision nanomedicine approaches. The colon cancer models demonstrated that HA-functionalized drug delivery platforms, which include liposomes and polymeric nanoparticles and micelles and nanogels and HA-drug conjugates, achieved better tumor accumulation and cell-specific uptake than their non-targeted counterparts. HA-based carriers which take

advantage of the tumor microenvironment show environment-responsive degradation because hyaluronidases get overexpressed in that space. HA-based carriers which take advantage of the tumor microenvironment show environment-responsive degradation because hyaluronidases get overexpressed in that space. HA-based carriers show environment-responsive degradation which enables them to release drugs at specific locations in a controlled manner. HA-targeted therapy enhances the cytotoxic effectiveness of chemotherapeutic agents while providing an effective method to treat multidrug resistance (MDR). HA-mediated targeting enables drug retention within CD44-positive colon cancer cells which express high amounts of P-glycoprotein (P-gp) drug efflux transporters. The treatment uses HA to target cancer cells which leads to increased intracellular drug retention and better treatment results in resistant cancer cell populations.(de Paula et al., 2023)

HA-based systems have been used to deliver chemotherapeutic drugs together with P-gp inhibitors and siRNA and immune-modulating agents which enhance anti-tumor effects through multiple treatment methods. Preclinical studies demonstrate that HA-targeted therapeutics provide reduced systemic toxicity and better pharmacokinetic performance and increased tumor penetration and more effective tumor growth control. HA functions as a strong biological ligand which boosts target accuracy and treatment effectiveness and provides new combination methods for successful colon cancer treatment.(W. Li et al., 2023)

4.3 Advantages of HA for receptor-mediated uptake

Hyaluronic acid (HA) provides multiple benefits which enable receptor-based uptake. This quality makes HA the most frequently used targeting ligand for nanomedicine applications in colon cancer research. The main benefit of this system lies in its ability to bind specifically to CD44 receptors which exist in high quantities on colon cancer cells and their metastatic tumors and cancer stem cells (CSCs). The system guarantees that HA-functionalized nanocarriers will be internally absorbed through CD44-linked endocytosis which leads to greater drug concentration in tumor cells while protecting healthy cells from drug exposure. HA demonstrates safe use in systemic delivery because of its biocompatibility and biodegradable properties and its lack of immunogenic effects which also decrease the chance of negative side effects. The hydrophilic polymeric backbone of the material enhances both aqueous solubility and circulation stability and dispersibility of drug-loaded nanoparticles.(Wickens et al., 2017)The enzyme-responsive behavior of hyaluronidases which are produced at excessive levels within tumor environments represents a primary benefit of HA. The HA degradation process begins when hyaluronidases which have been overexpressed in tumor microenvironments start to break down HA, which leads to drug release at specific locations and better treatment outcomes. The reactive functional groups of HA which include carboxyl groups and hydroxyl groups and N-acetyl moieties

enable diverse chemical transformations that permit drug attachment and imaging agent binding and targeting ligand connection and P-gp inhibitor partnership without losing their biorecognition capacity. HA enables deep tumor invasion because it binds to tumor matrices which have high CD44 expression. HA-modified nanoparticles increase CSC uptake while solving major challenges linked to multidrug resistance. The combined benefits of HA create an optimal targeting ligand which improves receptor-mediated intake while enhancing treatment efficiency and developing better drug absorption and effectiveness profiles for cancer treatments used in colon cancer treatment.(Al Jayoush et al., 2025)

5. P-gp Inhibitors in Colon Cancer Therapy

5.1 Classification of P-gp inhibitors (first, second, third generation)

Three generations of P-glycoprotein P-gp inhibitors have developed through three successive stages which brought major progress until the current time which shows consecutive improvements in their ability to fight cancer and their precise targeting of P-glycoprotein for clinical use against multidrug resistance in colon cancer and other types of cancer. The first-generation P-gp inhibitors which include verapamil cyclosporine A and quinidine received initial approval for treatment of non-oncologic conditions but researchers discovered their P-gp inhibitory effects at high dose levels. The drugs presented challenges in their clinical application because they lacked effective power and specific effects while creating high risks for systemic toxicity because of their interactions with chemotherapy medications. The second-generation inhibitors which include valspodar PSC-833 biricodar VX-710 and dexverapamil emerged from these constraints because they provided more effective P-glycoprotein P-gp blocking at lower toxic levels. The second-generation compounds maintained their ability to create major drug interactions because they blocked cytochrome P450 enzymes particularly CYP3A4 which led to unexpected alterations in chemotherapy drug absorption.To address these shortcomings, third-generation inhibitors such as tariquidar (XR9576), laniquidar, zosuquidar, and elacridar were engineered with high potency, superior selectivity toward P-gp, and minimal interference with metabolic enzymes. The inhibitors demonstrate strong reduction of target activity at nanomolar concentrations while showing effective results in two key areas which involve boosting drug levels inside cells and helping colon cancer models regain their ability to respond to chemotherapy. The use of these medications remains restricted because they have advanced to clinical trials yet they still exhibit problems with dangerous side effects and inconsistent treatment results and different reactions among patients. The third generation P-gp inhibitors represent the most suitable options for use in advanced drug delivery systems which develop HA-targeted nanocarriers that will work together to reverse multiple drug resistance in colon cancer treatment. The comparison between first generation

second generation and third generation P-gp inhibitors which includes their mechanisms and benefits and drawbacks has been compiled in Table 2.(Amin, 2013)

Table 2Classification of P-glycoprotein (P-gp) inhibitors with examples, mechanisms, advantages, and limitations.

Generation	Examples	Mechanism of Action	Advantages	Limitations	References
First Generation	Verapamil, Cyclosporine A	Weak competitive inhibition	Widely available	Low potency; drug–drug interactions; toxicity	(Amin, 2013)(Abdallah et al., 2015)
Second Generation	Valspodar (PSC833), Biricodar (VX-710), Dexverapamil	Improved affinity; partial CYP3A4 inhibition	Better potency vs first gen	Alters PK of chemotherapeutics; toxicity	(Loor, 1999)
(Loor, 1999)Third Generation	Tariquidar, Zosuquidar, Elacridar	High-affinity selective inhibition, minimal CYP effects	High potency; low toxicity	Limited clinical trial success	(H. Chen et al., 2017)

5.2 Mechanism of inhibition

P-glycoprotein (P-gp) inhibition can occur through several distinct mechanisms that collectively aim to reduce drug efflux, increase intracellular drug accumulation, and restore the cytotoxic effectiveness of chemotherapeutic agents in resistant colon cancer cells. The first mechanism involves competitive inhibition which functions when inhibitors bind to the drug-binding pocket of P-gp transmembrane domains to compete against chemotherapeutic substrates for their efflux. Non-competitive inhibition occurs when inhibitors bind to different allosteric sites on the transporter to create conformational changes which block ATP binding and ATP hydrolysis and substrate translocation without competing against drug substrates. ATPase inhibition occurs when P-gp inhibitors prevent ATP hydrolysis at the nucleotide-binding domains because this process

drives energy-based conformational shifts needed to transport drugs while blocking ATP turnover disables the efflux mechanism.Some inhibitors, particularly among second- and third-generation compounds, show their effects through two mechanisms which include changing membrane fluidity and modifying lipid microenvironment elements that lead to altered P-gp positioning and decreased P-gp function. The first group of agents works to downregulate P-gp expression through three methods which include blocking MDR1/ABCB1 gene transcription and causing mRNA destabilization and inducing proteasomal degradation of the transporter.(Amin, 2013)The research team uses advanced therapeutic methods which require them to stop all cell signaling processes that control P-gp protein production through three pathways that include PI3K/Akt and NF-κB and Wnt/β-catenin. The combined effects of these mechanisms create various ways to fight multidrug resistance which become more effective when used together with HA-modified nanoparticle systems that deliver targeted treatment.(Toffoli et al., 1996)

6.Synergistic Approach: HA with P-gp Inhibitors

6.1 Rationale for combination therapy: Hyaluronic Acid (HA) with P-gp Inhibitors

The use of hyaluronic acid (HA) for targeting P-glycoprotein (P-gp) while inhibiting its function creates an effective method to combat multidrug resistance (MDR) which occurs in colon cancer. HA specifically binds to the CD44 receptor, which is frequently overexpressed not only on bulk tumor cells but also on cancer stem cells (CSCs), the subpopulation responsible for tumor recurrence, metastasis, and intrinsic chemoresistance. The targeted interaction between HA and drug delivery systems that use HA as a functional component allows these systems to achieve selective drug accumulation in CD44-positive tumor areas while their drugs enter cells through receptor-mediated endocytosis, which leads to substantial drug absorption. The P-glycoprotein efflux system of tumor cells results in rapid removal of chemotherapeutic drugs from their cellular environment, which prevents the establishment of proper drug concentrations within the cells. The simultaneous administration of P-glycoprotein inhibitors together with chemotherapeutic agents operates as a solution to this challenge because it stops the drug removal process from cells, which improves drug retention and bioavailability and increases the cytotoxic effects of chemotherapy drugs on cancer cells. The HA–P-gp inhibitor combination enhances therapeutic efficacy at multiple levels.(Luo et al., 2019)HA improves the three tumor-related processes of drug delivery because it enhances two separate processes of drug absorption. The chemical combination of P-gp inhibitors together with P-gp inhibitors blocks the process through which drugs should exit the body while they stop multi-drug resistance. The tumor tissue at which HA shows better collection capacity functions as a drug delivery system because hyaluronidase enzymes break down HA to provide localized treatment which decreases the body-wide effects of P-gp inhibitors. The dual-target method

prevents CD44-overexpressing cells from using biological feedback systems which depend on CD44–HA signals to activate P-gp and other survival mechanisms. The combination of HA-based drug delivery and P-gp blocking creates a unified method which boosts drug delivery results and improves treatment outcomes while tackling both inner cell mechanisms and outer environment factors which cause chemotherapy resistance in colon cancer.(Xiao et al., 2015)

6.2 Targeting CD44-positive colon cancer cells

Targeting CD44-positive colon cancer cells has emerged as a highly effective therapeutic strategy because CD44 functions not only as a surface biomarker but also as a key driver of tumor progression, metastasis, and chemoresistance. Colorectal cancer cells show high CD44 expression which reaches its highest levels in cancer stem cell (CSC) populations that maintain their ability to self-renew and initiate tumors while resisting chemotherapy treatments. The CD44-high CSCs of tumors create a direct pathway to recurring tumors and spreading metastases which defines them as essential targets for treatment development. Hyaluronic acid (HA), the natural ligand for CD44, enables precise targeting through specific HA–CD44 interactions, which enable receptor-mediated endocytosis to deliver therapeutic agents directly into cells.(Zhu et al., 2022)The ligand-receptor specificity system results in enhanced drug penetration through CD44-overexpressing tumor areas while achieving reduced off-target toxic effects. CD44 binding activates cell signaling pathways that include Wnt/β-catenin and PI3K/Akt and MAPK/ERK and Rho GTPases which support tumor cell survival and epithelial-mesenchymal transition and drug resistance development. HA-based systems deliver drugs and gene silencers and P-glycoprotein P-gp inhibitors to CD44-positive cells which enables the systems to interrupt oncogenic pathways and make resistant cells more vulnerable to chemotherapy. The practice of targeting CD44-positive populations enables the destruction of CSCs which results in decreased relapse rates and enhanced long-term therapeutic results.(Ejima et al., 2023)CD44-targeted therapy creates a strong treatment platform for overcoming multiple drug resistance which leads to longer-lasting colon cancer treatment results when combined with nanocarriers and HA-P-gp inhibitor combination methods(Tang et al., 2025).

6.3 Overcoming multidrug resistance via dual approach

The combined dual treatment method which uses hyaluronic acid (HA) for CD44 targeting and P-glycoprotein (P-gp) inhibition creates an effective treatment method which helps to defeat multidrug resistance (MDR) in colon cancer. The cancer stem cells (CSCs) that express CD44 together with P-gp overexpression create two major factors which make chemotherapy treatment less effective in patients with multidrug resistance (MDR). The natural ligand HA binds to CD44 which allows scientists to target CD44-overexpressing tumor cells and

CSC subpopulations with high accuracy while the binding process enhances drug absorption through receptor-mediated endocytosis. The P-gp efflux pumps present in cancer cells will remove drugs from the cells even when the drugs have successfully entered them which results in keeping drug concentrations below effective levels. The co-delivery system which uses P-gp inhibitors solves this problem because it prevents drug release through ATP-dependent mechanisms while it enables drugs to stay inside cells. The HA-modified nanoparticles deliver this dual system which generates a combined treatment effect because HA provides accurate drug delivery to tumors while P-gp inhibitors stop drug release and return chemosensitivity to enhance drug killing effects(Gong et al., 2023).

CD44–HA interactions lead to P-gp expression upregulation through PI3K/Akt and Wnt/β-catenin signaling pathways. Therefore, simultaneous blockade of both CD44-mediated signaling and P-gp function disrupts essential survival pathways which enable chemoresistance(Espejo-Román et al., 2022). This dual-targeted approach ensures deeper tumor penetration, greater drug accumulation in resistant cell populations, and improved apoptotic response. Collectively, combining HA-based targeting with P-gp inhibition represents a mechanistically integrated and highly effective strategy to reverse MDR, reduce tumor recurrence, and enhance the overall therapeutic efficacy of colon cancer treatment.(Chaudhry et al., 2021)

7. Preclinical and Clinical Studies

7.1 In vitro studies on HA–P-gp inhibitor systems

In vitro research exploring HA–P-glycoprotein inhibitor systems have consistently yielded improvement in drug accumulation, cytotoxicity, and reversing multidrug resistance in colon cancer cell lines. Most of the studies employ CD44-overexpressing colon cancer model cells, like HCT116, HT-29, SW480, and LoVo cells, which also overexpress P-glycoprotein, allowing for testing of dual-target systems(L. Liu et al., 2023). HA-derived nanoparticles loaded with chemotherapeutics (doxorubicin, paclitaxel, 5-fluorouracil, and irinotecan) have demonstrated significant improvement in cellular uptake compared to non-targeted formulations, due to CD44-mediated endocytosis. When used as part of delivery with P-glycoprotein inhibitors (i.e., tariquidar, verapamil, cyclosporine A, or third-generation inhibitors), the HA–P-glycoprotein systems have shown to significantly diminish the rate of drug efflux, thereby improving intracellular drug retention. In multi-studies, intracellular fluorescence imaging and flow-cytometry has shown higher drug accumulation, and assays such as MTT, CCK-8, and apoptosis analysis has shown significant improvements in drug-induced cytotoxicities and percentages of induced apoptosis in resistant colon cancer cells.(Manickasamy et al., 2024) The HA-P-glycoprotein platforms have also been shown to downregulated MDR1 gene expression, inhibit ATPase

activity, and interrupt downstream signalling cascades contributing to chemoresistance. Importantly, the HA nanoparticles are also shown to be selectively toxic in CD44-high populations, while sparing normal colon epithelial cells, indicating a concerted effort of selective cytotoxicity at the cellular level. A comprehensive overview of key in vitro and in vivo findings demonstrating the therapeutic efficacy of HA–P-gp inhibitor combination systems is presented in Table 3.

Table 3 In vitro and in vivo outcomes of HA–P-gp inhibitor combination systems in colon cancer.

Study Model	**HA-Based Formulation**	**Therapeutic Agent(s)**	**Outcome Measured**	**Major Findings**	**Reference**
HCT116 (in vitro)	HA–polymeric NP	Doxorubicin + Tariquidar	Drug uptake, apoptosis	5-fold ↑ intracellular DOX; ↓ efflux; ↑ apoptosis rate	(Fox et al., 2015)(Xia et al., 2018)
HT-29 (in vitro)	HA–liposomes	Irinotecan + Verapamil	Cell viability, ROS generation	Significant ↓ cell viability; ↑ ROS; effective P-gp inhibition	(Mansoori et al., 2020)
SW480 (in vitro)	HA–micelles	Paclitaxel + P-gp inhibitor	Cytotoxicity, MDR1 expression	Strong reduction in MDR1; 3× ↑ drug accumulation	(Syed et al., 2017)
LoVo (in vitro)	HA–drug conjugate	HA–SN38 + Elacridar	Cell cycle arrest, apoptosis	Enhanced G2/M arrest; higher apoptosis vs. free drugs	(Qi et al., 2024)
Caco-2 (in vitro)	HA–nanogels	DOX + siRNA (MDR1)	Gene silencing, permeability	70–80% MDR1 silencing; ↑ permeability across cell layer	(Charbe et al., 2020)

Spheroid model (3D colon tumor)	HA–MSN	Oxaliplatin + Tariquidar	Penetration depth, cell death	Deep spheroid penetration; significant core necrosis	(Mirdamadian et al., 2022)
HCT116 xenograft (mouse)	HA–PLGA NP	DOX + Verapamil	Tumor volume, drug accumulation	4× ↑ tumor DOX concentration; strong tumor regression	(Abo Mansour et al., 2020)
SW620 xenograft (mouse)	HA–liposomes	Irinotecan + Tariquidar	Survival, systemic toxicity	↑ survival; ↓ systemic toxicity vs free drug	(Y. Liu et al., 2022)
HT-29 xenograft (mouse)	HA–micelles	Curcumin + P-gp inhibitor	Anti-inflammatory response	Significant ↓ NF-κB; ↓ tumor-associated inflammation	(Wahnou et al., 2025)
Patient-derived xenograft (PDX)	HA–NP co-loaded	Chemotherapy drug + P-gp blocker	Biodistribution, CSC reduction	Preferential tumor accumulation; ↓ CD44+/P-gp+ CSCs	(Gu et al., 2025)
Liver metastasis model	HA–NP	DOX + inhibitor	Metastasis suppression	Reduced hepatic metastasis nodules; ↓ EMT markers	(Kundu et al., 2013)
Orthotopic colon cancer model	HA–nanogel	Gene therapy + chemo	Tumor progression	Strong suppression of primary and metastatic tumor growth	(Attama et al., 2022)(Tseng et al., 2007)

7.2 In vivo models and outcomes

Animal studies assessing HA–P-gp inhibitor cytotoxic combination systems have provided compelling evidence for their increased therapeutic efficacy, modified pharmacokinetics, and multidrug resistance reversal potential in the treatment of colon cancer. Mouse xenografts are a widely used approach for evaluation of HA–P-gp inhibitor cytotoxic combination systems, most commonly utilizing CD44-overexpressing colon cancer cell lines, such as HCT116, SW620, or HT-29, or clinical patient-derived xenografts (PDXs).(Georges et al., 2019) HA-functionalized nanoparticles formulated with chemotherapeutics exhibit improved tumor accumulation, due to the enhanced permeability and retention (EPR) effects, and active CD44-mediated targeting. When co-formulated or co-administered with P-gp inhibitors, these combination systems exhibit enhanced intratumoral drug concentration and drug efflux reduction, resulting in enhanced tumor growth inhibition compared to standard chemotherapy or single agent formulations.(Yamamoto et al., 2019) In vivo imaging and biodistribution studies show prolonged circulation times with limited off-target organ deposition and selective delivery to tumor tissue. Furthermore, HA-P-gp inhibitor systems were found to downregulate MDR1/P-gp expression in tumors, cause higher total rates of apoptosis, greatly diminish CSC populations, and suppress metastatic spread to the liver and lung. Furthermore, the combination platforms have acceptable safety profiles with little systemic toxicity and have less cardiotoxic and gastrointestinal side effects than using free P-gp inhibitors, which is attributed to controlled release and targeting the tumor itself. Overall, the in vivo findings strongly support HA-based CD44 targeting combined with inhibition of P-gp displays great potential, which holds promise as a future therapeutic platform to treat drug-resistant colon cancer.(Spartalis et al., 2019)

7.3 Translational potential and current limitations

The clinical viability of HA–P-gp inhibitor combination systems for treating colon cancer is substantial, as they effectively address two critical factors in oncology: limited tumor specificity and multidrug resistance. HA-based nanocarriers enable selective targeting of tumor cells and cancer stem cell populations due to the natural affinity of HA for cancer antigen CD44, resulting in reduced off-target toxicity. At the same time, P-gp inhibitors restore intracellular drug retention and stimulate the therapeutic effects of conventional chemotherapeutics that are typically ‘sequestered through drug-efflux pumps. In addition, the platform is amenable towards creating dual-target systems, allowing for imaging agents, siRNA, CRISPR components, or immunomodulators towards creating multi-functional theragnostic platforms for personalized medicines to be incorporated.(Lou et al., 2025) However, strong preclinical evidence exists, and various limitations create barriers to translation into the clinic. The first barrier is the variable expression of CD44 in the patient population and the tumor subtyping, which could impede targeting efficacy. Furthermore, tumor heterogeneity

in penetrance, clearance from the mononuclear phagocyte system, and varying levels of hyaluronidase in tumor tissue could reduce HA uptake and degradation perspectives. Finally, even when loaded within a nanocarrier, systemic administration of P-gp inhibitors raises concerns regarding potential toxicity, drug interactions in an off-target manner, and drug pharmacokinetics particularly leading to concerns regarding altered bioavailability and coadministration with therapies.Challenges are also associated with large-scale reproducibility, batch-to-batch consistency, and regulatory approval pathways for complex nano-formulations.(Dey et al., 2023) Moreover, the majority of studies available are in pre-clinical space with limited clinical evidence (in humans) to confirm safety, optimal dose strategies, and long-term therapeutic benefits.

8. Challenges and Future Perspectives

The therapeutic effects of HA-mediated CD44 targeting combined with P-gp inhibition show promise for treatment development but face major obstacles in becoming usable medical practices. The safety and toxicity assessment process will require assessment because P-gp inhibitors which exist in HA-modified nanocarrier products or other forms present risks of off-target effects and cardiotoxicity and immunosuppression and altered pharmacokinetics in systemic administration. HA-based systems have the potential to trigger immune reactions or deposit in organs including the liver and spleen which requires testing of biocompatibility and safety through extended research that will study the targeted patient group. Drug delivery complications, such as heterogeneous expression of CD44, poor penetration of nanoparticles into the dense tumor stroma, or rapid opsonization can complicate treatment efficacy in HA-mediated targeting and release of drugs. The upcoming development stage needs to establish methods for enhancing tumor penetration and maintaining systemic stability and achieving endosomal entrapment. From a translational perspective, developing HA-P-gp inhibitor systems could provide a considerable opportunity for personalized medicine, as patient-specific biomarkers (e.g., levels of CD44 expression, MDR1 gene polymorphisms, intra-tumoral heterogeneity, and microenvironmental enzymatic activity) could be used for treatment selection and dose refinement.Utilizing personalized nanomedicine approaches, bolstered through molecular profiling, could improve predictability of therapy and diminish toxicity. In the future, HA–P-gp inhibitor nanomedicine will seek to develop multi-functional smart nanocarriers capable of co-delivery of chemotherapeutics, gene silencers (siRNA, CRISPR), immunotherapeutics, or materials that respond to the microenvironment for sequential or synergistic release. Advancements in biomaterial engineering, artificial intelligence–assisted design of nanoparticles and tumor-on-chip models will further enhance optimization. Lastly, regulatory standardization, large-scale reproducible techniques, and early-phase clinical trials will be necessary to confirm safety and efficacy. Altogether,

overcoming these hurdles while advancing new technologies will be necessary for moving forward HA–P-gp inhibitor–based nanomedicine towards clinical use in drug-resistant colon cancer.

9. Conclusion

In a promising approach against multidrug resistance (MDR) in colon cancer, hyaluronic acid (HA)-based CD44 targeting can be used jointly with P-glycoprotein (P-gp) inhibition in a potentially successful and mechanistically synergistic strategy. HA, a natural CD44 ligand, ensures targeted and efficient therapeutic delivery to CD44-overexpressing tumor cells and cancer stem-like cells by increasing receptor-mediated endocytosis and transportation into tumors. Besides, importantly, P-gp inhibition promotes the retention of chemotherapeutics in the cytoplasm by decreasing ATP-dependent efflux, which allows intracellular drug concentrations to rise and restores chemotherapy responses. Ultimately, both strategies counteract the chemosensitivity-inhibiting effects of both cells and tumor microenvironments, resulting in greater therapeutic potential than either approach alone. Preclinical studies have shown that HA-P-gp inhibitor systems may address these issues related to drug delivery, i.e., tumor size and burden, downregulating MDR pathophysiology, and minimized toxicity during chemotherapeutic use, have all been demonstrated in vitro and in vivo to be effective outcomes of HA-P-gp inhibitory systems. Clinically, however, obstacles to translation remain in the areas of overall toxicity, tumor heterogeneity, penetration distance, variable CD44 and P-gp expression and nanomedicine regulatory issues, clinical barriers remain.In the future, research should focus on: patient-specific strategies (individualism), next-generation biomaterial designs for innovative therapies, expertly designed clinical trials to verify efficacy and long-term safety, and multi-functional nanotrophics. On the whole, HA-based CD44-targeted delivery plus P-gp inhibition are an attractive and innovative next step in developing therapies for enhanced outcomes in drug-resistant colon cancer.

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